

## Successful and Safe Treatment of Intestinal Graft-Versus-Host Disease (GvHD) with Pooled- Donor Full Ecosystem Microbiota Biotherapeutics

Florent Malard, MD PhD,<sup>\*,1</sup> Faezeh Legrand, MD,<sup>\*,2</sup> Jérôme Cornillon, MD,<sup>\*,3</sup> Amandine Le Bourgeois, MD,<sup>\*,4</sup> Jean-Baptiste Mear, MD,<sup>\*,5</sup> Deborah Desmier, MD,<sup>\*,6</sup> Emilie Plantamura, PhD PharmD,<sup>\*,7</sup> Ronald Carter, MD MSc, MPH,<sup>\*,7</sup> Thierry Lamy De La Chapelle, MD PhD,<sup>\*,5</sup> Patrice Chevallier, MD,<sup>8</sup> Denis Guyotat, MD PhD,<sup>\*,3</sup> Didier Blaise, MD PhD,<sup>2</sup> Mohamad Mohty, MD PhD<sup>9</sup>

<sup>1</sup>Service d'Hématologie Clinique et de Thérapie Cellulaire, Hospital Saint Antoine, Paris, France

<sup>2</sup>Institut Paoli Calmettes, Marseille, France

<sup>3</sup>Hematology department, Lucien Neuwirth Cancer Institute, Saint-Priest-en-Jarez, France

<sup>4</sup>Service d'Hématologie, CHU Nantes, Nantes, France

<sup>5</sup>Clinical hematology, University Hospital of Rennes, Rennes, France

<sup>6</sup>Service d'Hématologie et de thérapie cellulaire, CHU de Poitiers, Poitiers, France

<sup>7</sup>MaaT Pharma, LYON, France

<sup>8</sup>Service d'Hématologie, CHU Nantes, Nantes, France

<sup>9</sup>Hôpital Saint-Antoine, Paris, France

*bloodjournal Blood blood* (2019) 134 (Supplement\_1) : 1993.

<http://doi.org/10.1182/blood-2019-125540>

### Introduction

Intestinal Graft-versus-Host Disease (GvHD), following allogeneic Hematopoietic Stem Cell Transplantation (allo-HSCT), comes with a high mortality rate and a reduced life-expectancy. In this context, failure to respond to steroid therapy is associated with an absence of further therapeutic options, thereby representing an unmet medical need. A marked reduction in gut microbiota diversity leading to loss of functions was strongly associated with reduced overall survival in GvHD, while a high gut microbiota diversity appears to be protective. Aiming at restoring microbiome functions, Fecal Microbiota Transfer proved to be a promising treatment modality in this challenging clinical situation, with recent studies reporting favorable results in steroid refractory-acute GVHD (SR-aGVHD) patients. Here we report

on the use of "MaaT013", a standardized, pooled-donor, high-richness microbiota biotherapeutic, used to treat 8 patients with intestinal-predominant aGvHD.

## **Patients and methods**

Eight allo-HSCT recipients with steroid-dependent or steroid-refractory gastrointestinal GvHD (classical aGvHD n=3, late-onset aGvHD n=2; aGvHD with overlap syndrome n=3) were treated with the MaaT013 biotherapeutic as part of a compassionate use program. These patients had previously received and failed 1 to 5 lines (median 2.5) of GvHD systemic treatments. MaaT013 microbiota biotherapeutics were provided as a pharmaceutical preparation to hospitals by the developer, "MaaT Pharma". Each patient received 1 to 3 doses (median: 3; total doses administered: 21) of MaaT013, a 150 mL bag, by enema (n=7) or nasogastric tube (n=1). GvHD response was evaluated 7 days after each administration and 28 days after the first dose.

Prepared under Good Manufacturing Practices, MaaT013 biotherapeutics are characterized by a highly consistent richness of 455 +/- 3% Operational Taxonomic Units (OTUs) and an Inverse Simpson index greater than 20. Batch release specifications are based on potency (viability), identity (diversity), and purity (microbiological safety testing following regulatory guidelines and proportion of proinflammatory species), ensuring the desired consistency between batches.

## **Results**

We observed 6/8 positive responses at D28 after first dosing, including 3 Complete Responses (CR), one Very Good Partial Response (VGPR), and 2 Partial Responses (PR). Considering the best GI response achieved, all (8/8) patients experienced at least a PR, with 3 CRs, 2 VGPRs and 3 PRs. The 3 patients with CR were still alive at last follow-up (60 to 192 days after last dosing; median: 115 days) and were able to taper and stop steroids and immunosuppressants without relapse of GI symptoms. Of note, among these 3 patients, mild mouth chronic GvHD symptoms persisted in one patient, and relapse of hematologic malignancy was observed in another patient. Among the 8 treated patients, 5 were still alive at last follow-up (60 to 232 days after last dosing; median: 125 days).

The safety of the MaaT013 microbiota biotherapeutic was satisfactory in all patients. One patient developed a possibly related sepsis one day after the third MaaT013 administration. In the latter case, no pathogen was identified in blood cultures, and the patient recovered after a course of antibiotics.

## **Conclusions**

We herein report for the first time the treatment of 8 patients with steroid-dependent or steroid-refractory intestinal aGvHD using a full ecosystem, standardized, pooled-donor, high-richness biotherapeutic.

Overall, 3/8 patients attained a complete response following treatment with the off-the-shelf MaaT013 product, shown to be safe and effective in these immunocompromised patients with severe conditions, warranting further exploration of the full ecosystem microbiota restoration approach.

## Disclosures

**Malard:** *Sanofi*: Honoraria; *JAZZ pharmaceutical*: Honoraria; *Astellas*: Honoraria; *Therakos*/  
*Mallinckrodt*: Honoraria; *Keocyte*: Honoraria; *Janssen*: Honoraria. **Plantamura:** *MaaT Pharma*:  
Employment. **Carter:** *MaaT Pharma*: Employment. **Chevallier:** *Jazz Pharmaceuticals*: Honoraria; *Incyte*:  
Consultancy, Honoraria; *Daiichi Sankyo*: Honoraria. **Blaise:** *Pierre Fabre medicaments*: Honoraria;  
*Molmed*: Consultancy, Honoraria; *Jazz Pharmaceuticals*: Honoraria; *Sanofi*: Honoraria. **Mohty:** *Jazz*  
*Pharmaceuticals*: Honoraria, Research Funding.

## Author notes

\*Asterisk with author names denotes non-ASH members.